Lesson 1 · Spot the AI around you · for your pupils

Learns or follows a rule?

40 minutes | Ages 10–16 · adapt for other ages below | No electricity needed | No devices needed | Photocopy-friendly

Big question
How can we tell if a machine uses AI?

Pupils will be able to...

- Say what AI is in their own words.
- Sort everyday tools into “learns from examples” and “follows a rule”, and explain why.
- Give one example of AI from their own life.

You need

- Board and chalk (or paper on the wall)
- 8 picture cards — draw them on scrap paper or simply call out the names
- Worksheet: AI detective (one per pair, optional)

Before class

Draw two large boxes on the board and label them “Learns from examples” and “Follows a rule”. Prepare 8 cards: calculator, voice typing, school bell, spam filter, solar lamp timer, phone keyboard, flood alert, traffic light.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Ask how a new pupil learns everyone’s names. Take 3–4 answers.	*“Did someone give you a rule for every face, or did you learn by seeing people many times?”*	Share ideas; notice that we learn from examples.
Teach 5–13 min	Draw a solar lamp and a clock: “6 pm → on”. Then draw a phone keyboard suggesting a word. Write the one-sentence definition of AI on the board.	*“The lamp follows a rule a person wrote. The keyboard learned from lots of examples, so it can make a best guess. That is AI.”*	Copy the definition; repeat it to a partner in their own words.
Sort it 13–25 min	Hold up or call out each card. Pupils vote: left hand up for “follows a rule”, right hand for “learns from examples”. Place the card in a box and ask one pupil to explain.	*“Why do you think so? What examples would it need to learn from?”*	Vote, explain, and change their minds if someone gives a good reason.
Pair talk 25–33 min	Pairs think of one more thing at home, at school or in the market that might use AI. Collect three ideas and let the class decide which box.	*“Does it learn from examples, or does it only follow steps someone set?”*	Discuss in pairs; share one idea.
Wrap-up 33–40 min	Exit ticket: each pupil writes or says one sentence starting “AI is...” and one example.	*“Tomorrow, look for one more example on the way home.”*	Complete the exit ticket.

Check understanding
Listen for “learns from examples” or “makes a guess”. If pupils say “it works by itself”, ask: “Did it learn, or did someone set it?”

Exit ticket
AI is… (one sentence) + one example from my life.

Common mix-ups

Pupils may think…	Help them see…
Automatic means AI.	Ask who decided what the machine does: a person’s rule, or patterns learned from examples?
AI is a robot.	Point to the phone keyboard example — AI is often invisible software.
AI is always right.	Remind pupils the keyboard sometimes suggests the wrong word: it is a best guess.

Adapt it

Younger pupils
Ages 6–9: use pictures only and play “teacher robot”. First act as a machine that follows one rule, then as one that copies what pupils show you.

Older pupils and university
Ask what examples a spam filter would need to learn from, and introduce the words “training data” and “model”.

Large classes
Vote by standing up or sitting down instead of raising hands; let row leaders explain the class’s choice.

No board or no space
Use two corners of the room as the boxes, or draw them on the ground outside with a stick.

Short on time? Do the warm-up, the teach step and six cards of the sort (20 minutes).

What you need to know

Some machines only follow fixed steps that people write. Most AI today is different: it learns patterns from many examples, then makes a best guess.

Artificial intelligence (AI)
A computer system that works out an output — a prediction, suggestion, decision or new text — from what it is given. Most AI today learns this from examples.

Pattern
Something that repeats in many examples — for example, “results are…” is very often followed by “ready” or “out”.

Rule
A fixed “if this, then that” instruction a person writes for a machine, such as “at 6 pm, switch on”. The machine follows it the same way every time.

Prediction
The system’s best guess about something it can’t know for sure — “this message is spam”, “this river will flood”. It can be about now, and it can be wrong.

My notes — how did it go?